1. Introduction

The goal of this project is to build a machine learning model that predicts house prices using linear regression.
The project demonstrates the complete machine learning workflow including data loading, exploration, preprocessing, model training, and evaluation.

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2. Dataset Description

The California Housing dataset contains information about housing areas in California.

Features include:

- Median income
- House age
- Average rooms
- Population
- Location
- House value (target)

Total samples: **20,640**
Total features: **8**

The target variable is **MedHouseVal** (house price).

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3. Methodology

Step 1: Environment Setup

Installed required libraries:

- pandas
- numpy
- scikit-learn
- matplotlib
- seaborn

Step 2: Data Loading

Loaded dataset using sklearn and converted it into a pandas DataFrame.

Step 3: Data Exploration

- Checked dataset shape and info
- Generated heatmap
- Observed correlation between features and price
- Median income showed strongest relationship with price.

Step 4: Data Preparation

Separated:

- Features (X)
- Target (y)

Split dataset:

- 80% training
- 20% testing

Step 5: Model Training

Used Linear Regression model from sklearn.

Model was trained using training data.

Step 6: Prediction

Predicted house prices on test data.

Step 7: Evaluation

Used metrics:

- MAE
- RMSE
- R^2 Score

🔹 4. Results

(Write your actual output values here)

Example:

- MAE: 0.53
- RMSE: 0.74

- R^2 Score: 0.64

The R^2 score indicates the model predicts house prices with reasonable accuracy.

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5. Visualization

A scatter plot of actual vs predicted values shows that predictions are close to actual values, indicating a good model fit.

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6. Conclusion

The linear regression model successfully predicts house prices with moderate accuracy.
Median income was found to be the most influential feature.

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7. Future Improvements

- Use Ridge or Lasso regression
- Feature engineering
- Try polynomial regression
- Use more complex models